

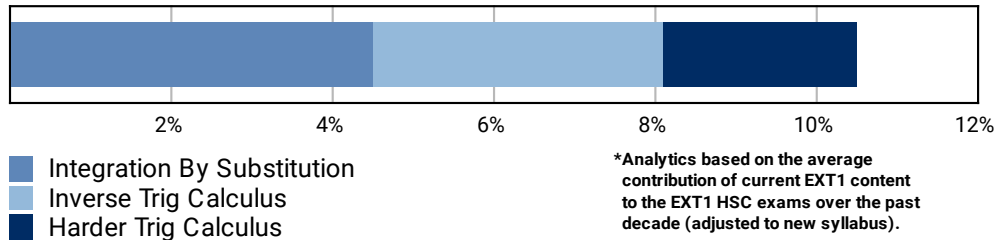
EXT 1: Calculus (Ext1), C2 Further Calculus Skills (Y12)

Integration By Substitution (Ext1)

Teacher: Ljiljana Todic

Exam Equivalent Time: 91.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

C2 Further Calculus Skills



HISTORICAL CONTRIBUTION

- C2 Further Calculus Skills has a past contribution of 10.5% per HSC exam over the past 10 years.
- This topic has been split into three sub-categories for analysis purposes which are: 1- Integration By Substitution (4.5%), 2-Inverse Trig Calculus (3.6%) and 3-Harder Trig Calculus (2.4%).
- This analysis looks at Integration By Substitution.

HSC ANALYSIS - What to expect and common pitfalls

- Integration by Substitution (4.5%) is a super consistent topic area, asked in a dedicated 3-mark question every year of the last decade, with the exception of 2020 where a 5-mark allocation required students to differentiate and then integrate.
- Approach this area by concentrating on making no mistakes and developing speed, in that order!
- Pitfalls:** One of the few problem areas flagged by markers has been when definite integral limits look inverted after the substitution (see 2007 Ext1 1e).

Questions

1. Calculus, EXT1 C2 SM-Bank 1 MC

With a suitable substitution, $\int_1^5 (2x-1)\sqrt{2x+1} dx$ can be expressed as

- A. $\frac{1}{2} \int_1^5 \left(u^{\frac{3}{2}} + u^{\frac{1}{2}}\right) du$
- B. $2 \int_3^{11} \left(u^{\frac{3}{2}} + u^{\frac{1}{2}}\right) du$
- C. $2 \int_1^5 \left(u^{\frac{3}{2}} - 2u^{\frac{1}{2}}\right) du$
- D. $\frac{1}{2} \int_3^{11} \left(u^{\frac{3}{2}} - 2u^{\frac{1}{2}}\right) du$

2. Calculus, EXT1 C2 2013 HSC 5 MC

Which integral is obtained when the substitution $u = 1 + 2x$ is applied to $\int x\sqrt{1+2x} dx$?

- (A) $\frac{1}{4} \int (u-1)\sqrt{u} du$
- (B) $\frac{1}{2} \int (u-1)\sqrt{u} du$
- (C) $\int (u-1)\sqrt{u} du$
- (D) $2 \int (u-1)\sqrt{u} du$

3. Calculus, EXT1 C2 2020 SPEC1 2

Evaluate $\int_{-1}^0 \frac{1+x}{\sqrt{1-x}} dx$, using the substitution $u = 1-x$. (3 marks)

4. Calculus, EXT1 C2 2021 HSC 11c

Use the substitution $u = x+1$ to find $\int x\sqrt{x+1} dx$. (3 marks)

5. Calculus, EXT1 C2 2004 HSC 1e

Use the substitution $u = x - 3$ to evaluate

$$\int_3^4 x\sqrt{x-3} \, dx. \quad (3 \text{ marks})$$

6. Calculus, EXT1 C2 2006 HSC 1b

Using the substitution $u = x^4 + 8$, or otherwise, find

$$\int x^3\sqrt{x^4+8} \, dx. \quad (3 \text{ marks})$$

7. Calculus, EXT1 C2 2010 HSC 1e

Use the substitution $u = 1 - x$ to evaluate $\int_0^1 x\sqrt{1-x} \, dx. \quad (3 \text{ marks})$

8. Calculus, EXT1 C2 2011 HSC 1d

Using the substitution $u = \sqrt{x}$, evaluate $\int_1^4 \frac{e^{\sqrt{x}}}{\sqrt{x}} \, dx. \quad (3 \text{ marks})$

9. Calculus, EXT1 C2 2012 HSC 11d

Use the substitution $u = 2 - x$ to evaluate $\int_1^2 x(2-x)^5 \, dx. \quad (3 \text{ marks})$

10. Calculus, EXT1 C2 2015 HSC 11e

Use the substitution $u = 2x - 1$ to evaluate $\int_1^2 \frac{x}{(2x-1)^2} \, dx. \quad (3 \text{ marks})$

11. Calculus, EXT1 C2 2016 HSC 11b

Use the substitution $u = x - 4$ to find $\int x\sqrt{x-4} \, dx. \quad (3 \text{ marks})$

12. Calculus, EXT1 C2 2017 HSC 11e

Evaluate $\int_0^3 \frac{x}{\sqrt{x+1}} \, dx$, using the substitution $x = u^2 - 1. \quad (3 \text{ marks})$

13. Calculus, EXT1 C2 2018 HSC 11f

Evaluate $\int_{-3}^0 \frac{x}{\sqrt{1-x}} \, dx$, using the substitution $u = 1 - x. \quad (3 \text{ marks})$

14. Calculus, EXT1 C2 2019 HSC 13a

Use the substitution $u = \cos^2 x$ to evaluate $\int_0^{\frac{\pi}{4}} \frac{\sin 2x}{4 + \cos^2 x} \, dx. \quad (3 \text{ marks})$

15. Calculus, EXT1 C2 2020 HSC 13a

i. Find $\frac{d}{d\theta}(\sin^3 \theta). \quad (1 \text{ mark})$

ii. Use the substitution $x = \tan \theta$ to evaluate $\int_0^1 \frac{x^2}{(1+x^2)^{\frac{5}{2}}} \, dx. \quad (4 \text{ marks})$

16. Calculus, EXT1 C2 2009 HSC 1f

Using the substitution $u = x^3 + 1$, or otherwise, evaluate $\int_0^2 x^2 e^{x^3+1} \, dx. \quad (3 \text{ marks})$

17. Calculus, EXT1 C2 2005 HSC 1d

Using the substitution $u = 2x^2 + 1$, or otherwise, find $\int x(2x^2 + 1)^{\frac{5}{4}} \, dx. \quad (3 \text{ marks})$

18. Calculus, EXT1 C2 2007 HSC 1e

Use the substitution $u = 25 - x^2$ to evaluate $\int_3^4 \frac{2x}{\sqrt{25-x^2}} \, dx. \quad (3 \text{ marks})$

19. Calculus, EXT1 C2 2008 HSC 2a

Use the substitution $u = \log_e x$ to evaluate $\int_e^{e^2} \frac{1}{x(\log_e x)^2} \, dx. \quad (3 \text{ marks})$

20. Calculus, EXT1 C2 2013 HSC 11f

Use the substitution $u = e^{3x}$ to evaluate $\int_0^{\frac{1}{3}} \frac{e^{3x}}{e^{6x} + 1} \, dx. \quad (3 \text{ marks})$

Evaluate $\int_2^5 \frac{x}{\sqrt{x-1}} dx$ using the substitution $x = u^2 + 1$. (3 marks)

Worked Solutions

1. Calculus, EXT1 C2 SM-Bank 1 MC

Let $u = 2x + 1 \Rightarrow u - 2 = 2x - 1$

$$\frac{du}{dx} = 2 \Rightarrow dx = \frac{1}{2} du$$

When $x = 5$, $u = 11$

When $x = 1$, $u = 3$

$$\therefore \int_1^5 (2x - 1)\sqrt{2x + 1} dx$$

$$= \frac{1}{2} \int_3^{11} (u - 2)u^{\frac{1}{2}} du$$

$$= \frac{1}{2} \int_3^{11} u^{\frac{3}{2}} - 2u^{\frac{1}{2}} du$$

$$\Rightarrow D$$

2. Calculus, EXT1 C2 2013 HSC 5 MC

Let $u = 1 + 2x$

$$\therefore x = \frac{1}{2}(u - 1)$$

$$\frac{du}{dx} = 2$$

$$\therefore dx = \frac{1}{2} du$$

$$\int x\sqrt{1+2x} dx$$

$$= \int \frac{1}{2}(u - 1) \times u^{\frac{1}{2}} \times \frac{1}{2} du$$

$$= \frac{1}{4} \int (u - 1)\sqrt{u} du$$

$$\Rightarrow A$$

3. Calculus, EXT1 C2 2020 SPEC1 2

$$u = 1 - x \Rightarrow x = 1 - u$$

$$\frac{du}{dx} = -1 \Rightarrow dx = -du$$

When $x = 0$, $u = 1$

$$x = -1, u = 2$$

$$\begin{aligned} \int_{-1}^0 \frac{1+x}{\sqrt{1-x}} dx &= - \int_2^1 \frac{2-u}{\sqrt{u}} du \\ &= \int_1^2 2u^{-\frac{1}{2}} - u^{\frac{1}{2}} du \\ &= \left[4u^{\frac{1}{2}} - \frac{2}{3}u^{\frac{3}{2}} \right]_1^2 \\ &= 4\sqrt{2} - \frac{4\sqrt{2}}{3} - \left(4 - \frac{2}{3} \right) \\ &= \frac{8\sqrt{2}}{3} - \frac{10}{3} \end{aligned}$$

4. Calculus, EXT1 C2 2021 HSC 11c

$$u = x + 1 \Rightarrow \frac{du}{dx} = 1$$

$$\begin{aligned} \int x\sqrt{x-1} dx &= \int (u-1)\sqrt{u} du \\ &= \int u^{\frac{3}{2}} - u^{\frac{1}{2}} du \\ &= \frac{2}{5}u^{\frac{5}{2}} - \frac{2}{3}u^{\frac{3}{2}} + c \\ &= \frac{2}{5}(x+1)^{\frac{5}{2}} - \frac{2}{3}(x+1)^{\frac{3}{2}} + c \end{aligned}$$

5. Calculus, EXT1 C2 2004 HSC 1e

Let $u = x - 3$

$$\Rightarrow x = u + 3$$

$$\frac{du}{dx} = 1$$

$$\Rightarrow dx = du$$

When $x = 4$, $u = 1$

$$x = 3, u = 0$$

$$\begin{aligned} \int_3^4 x\sqrt{x-3} dx &= \int_0^1 (u+3) u^{\frac{1}{2}} du \\ &= \int_0^1 u^{\frac{3}{2}} + 3u^{\frac{1}{2}} du \\ &= \left[\frac{2}{5}u^{\frac{5}{2}} + 3 \times \frac{2}{3}u^{\frac{3}{2}} \right]_0^1 \\ &= \left[\frac{2}{5}u^{\frac{5}{2}} + 2u^{\frac{3}{2}} \right]_0^1 \\ &= \left[\left(\frac{2}{5} + 2 \right) - 0 \right] \\ &= 2\frac{2}{5} \end{aligned}$$

6. Calculus, EXT1 C2 2006 HSC 1b

$$u = x^4 + 8$$

$$\frac{du}{dx} = 4x^3$$

$$\frac{1}{4} du = x^3 dx$$

$$\therefore \int x^3 \sqrt{x^4 + 8} dx$$

$$= \int u^{\frac{1}{2}} \cdot \frac{1}{4} \cdot du$$

$$= \frac{1}{4} \int u^{\frac{1}{2}} du$$

$$= \frac{1}{4} \cdot \frac{2}{3} \cdot u^{\frac{3}{2}} + c$$

$$= \frac{1}{6} u^{\frac{3}{2}} + c$$

$$= \frac{1}{6} (x^4 + 8)^{\frac{3}{2}} + c$$

7. Calculus, EXT1 C2 2010 HSC 1e

$$u = 1 - x \Rightarrow x = 1 - u$$

$$\frac{du}{dx} = -1 \Rightarrow du = -dx$$

$$\text{When } x = 1, u = 0$$

$$x = 0, u = 1$$

$$\therefore \int_0^1 x \sqrt{1-x} dx$$

$$= - \int_1^0 (1-u) u^{\frac{1}{2}} du$$

$$= \int_1^0 (u-1) u^{\frac{1}{2}} du$$

$$= \int_1^0 \left(u^{\frac{3}{2}} - u^{\frac{1}{2}} \right) du$$

$$= \left[\frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} \right]_1^0$$

$$= \left[0 - \left(\frac{2}{5} - \frac{2}{3} \right) \right]$$

$$= - \left(\frac{6}{15} - \frac{10}{15} \right)$$

$$= \frac{4}{15}$$

8. Calculus, EXT1 C2 2011 HSC 1d

$$u = \sqrt{x} = x^{\frac{1}{2}}$$

$$\frac{du}{dx} = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$$

$$du = \frac{dx}{2\sqrt{x}}$$

$$\therefore 2du = \frac{dx}{\sqrt{x}}$$

When $x = 4$, $u = 2$

$x = 1$, $u = 1$

$$\begin{aligned}\therefore \int_1^4 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx &= \int_1^2 e^u \times 2 du \\ &= 2[e^u]_1^2 \\ &= 2[e^2 - e^1] \\ &= 2e(e - 1)\end{aligned}$$

9. Calculus, EXT1 C2 2012 HSC 11d

$$u = 2 - x$$

$$\therefore x = 2 - u$$

$$\frac{du}{dx} = -1$$

$$\therefore dx = -du$$

When $x = 2$, $u = 0$

When $x = 1$, $u = 1$

$$\begin{aligned}\therefore \int_1^2 x(2-x)^5 dx &= \int_1^0 - (2-u)u^5 du \\ &= \int_1^0 u^6 - 2u^5 du \\ &= \left[\frac{1}{7}u^7 - \frac{2}{6}u^6 \right]_1^0 \\ &= \left[0 - \left(\frac{1}{7} - \frac{1}{3} \right) \right] \\ &= - \left(\frac{3}{21} - \frac{7}{21} \right) \\ &= \frac{4}{21}\end{aligned}$$

10. Calculus, EXT1 C2 2015 HSC 11e

$$u = 2x - 1$$

$$\Rightarrow 2x = u + 1$$

$$x = \frac{1}{2}(u + 1)$$

$$\frac{du}{dx} = 2$$

$$dx = \frac{du}{2}$$

When $x = 2$, $u = 3$

$x = 1$, $u = 1$

$$\begin{aligned} \therefore \int_1^2 \frac{x}{(2x-1)^2} dx &= \int_1^3 \frac{1}{2}(u+1) \cdot \frac{1}{u^2} \cdot \frac{du}{2} \\ &= \frac{1}{4} \int_1^3 \left(\frac{u+1}{u^2} \right) du \\ &= \frac{1}{4} \int_1^3 \frac{1}{u} + u^{-2} du \\ &= \frac{1}{4} [\ln u - u^{-1}]_1^3 \\ &= \frac{1}{4} \left[\left(\ln 3 - \frac{1}{3} \right) - (\ln 1 - 1) \right] \\ &= \frac{1}{4} \left(\ln 3 - \frac{1}{3} + 1 \right) \\ &= \frac{1}{4} \left(\ln 3 + \frac{2}{3} \right) \end{aligned}$$

11. Calculus, EXT1 C2 2016 HSC 11b

$$u = x - 4 \Rightarrow x = u + 4$$

$$\frac{du}{dx} = 1 \Rightarrow dx = du$$

$$\begin{aligned} \therefore \int x\sqrt{x-4} dx &= \int (u+4) \cdot u^{\frac{1}{2}} du \\ &= \int u^{\frac{3}{2}} + 4u^{\frac{1}{2}} du \\ &= \frac{2}{5} u^{\frac{5}{2}} + 4 \cdot \frac{2}{3} u^{\frac{3}{2}} + c \\ &= \frac{2}{5} (x-4)^{\frac{5}{2}} + \frac{8}{3} (x-4)^{\frac{3}{2}} + c \end{aligned}$$

12. Calculus, EXT1 C2 2017 HSC 11e

$$x = u^2 - 1$$

$$u^2 = x + 1$$

$$u = \sqrt{x+1}$$

$$du = \frac{1}{2\sqrt{x+1}} dx$$

If $x = 3$, $u = 2$

$x = 0$, $u = 1$

$$\begin{aligned} \therefore \int_0^3 \frac{x}{\sqrt{x+1}} dx &= 2 \int_1^2 u^2 - 1 du \\ &= 2 \left[\frac{u^3}{3} - u \right]_1^2 \\ &= 2 \left[\left(\frac{8}{3} - 2 \right) - \left(\frac{1}{3} - 1 \right) \right] \\ &= 2 \left(\frac{2}{3} + \frac{2}{3} \right) \\ &= \frac{8}{3} \end{aligned}$$

13. Calculus, EXT1 C2 2018 HSC 11f

$$u = 1 - x \Rightarrow x = 1 - u$$

$$\frac{du}{dx} = -1$$

$$dx = -du$$

$$\text{When } x = 0, u = 1$$

$$\text{When } x = -3, u = 4$$

$$\begin{aligned} \int_{-3}^0 \frac{x}{\sqrt{1-x}} dx &= - \int_4^1 \frac{1-u}{\sqrt{u}} du \\ &= - \int_4^1 u^{-\frac{1}{2}} - u^{\frac{1}{2}} du \\ &= - \left[2u^{\frac{1}{2}} - \frac{2}{3}u^{\frac{3}{2}} \right]_4^1 \\ &= - \left[\left(2 - \frac{2}{3} \right) - \left(2\sqrt{4} - \frac{2}{3}(\sqrt{4})^3 \right) \right] \\ &= - \left[\frac{4}{3} - \left(4 - \frac{16}{3} \right) \right] \\ &= -\frac{8}{3} \end{aligned}$$

14. Calculus, EXT1 C2 2019 HSC 13a

$$u = \cos^2 x$$

$$\frac{du}{dx} = -2 \sin x \cos x$$

$$= -\sin 2x$$

$$du = -\sin 2x dx$$

$$\text{When } x = \frac{\pi}{4}, u = \frac{1}{2}$$

$$\text{When } x = 0, u = 1$$

$$\begin{aligned} \therefore \int_0^{\frac{\pi}{4}} \frac{\sin 2x}{4 + \cos^2 x} dx &= - \int_1^{\frac{1}{2}} \frac{du}{4 + u} \\ &= - [\ln(4 + u)]_1^{\frac{1}{2}} \\ &= - (\ln 4.5 - \ln 5) \\ &= - \ln \frac{9}{10} \\ &= \ln \frac{10}{9} \end{aligned}$$

15. Calculus, EXT1 C2 2020 HSC 13a

i. $\frac{d}{d\theta}(\sin^3 \theta) = 3 \cos \theta \sin^2 \theta$

ii. Let $x = \tan \theta$

$$\frac{dx}{d\theta} = \sec^2 \theta \Rightarrow dx = \sec^2 \theta d\theta$$

When $x = 1$, $\theta = \frac{\pi}{4}$

When $x = 0$, $\theta = 0$

$$\begin{aligned} \int_0^1 \frac{x^2}{(1+x^2)^{\frac{5}{2}}} dx &= \int_0^{\frac{\pi}{4}} \frac{\tan^2 \theta}{(1+\tan^2 \theta)^{\frac{5}{2}}} \times \sec^2 \theta d\theta \\ &= \int_0^{\frac{\pi}{4}} \frac{\tan^2 \theta}{(\sec^2 \theta)^{\frac{5}{2}}} \times \sec^2 \theta d\theta \\ &= \int_0^{\frac{\pi}{4}} \frac{\sin^2 \theta}{\cos^2 \theta} \cdot \frac{1}{(\sec^2 \theta)^{\frac{3}{2}}} d\theta \\ &= \int_0^{\frac{\pi}{4}} \frac{\sin^2 \theta}{\cos^2 \theta} \cdot \frac{1}{\sec^3 \theta} d\theta \\ &= \int_0^{\frac{\pi}{4}} \sin^2 \theta \cos \theta d\theta \\ &= \frac{1}{3} [\sin^3 \theta]_0^{\frac{\pi}{4}} \\ &= \frac{1}{3} \left(\sin^3 \frac{\pi}{4} - 0 \right) \\ &= \frac{1}{3} \left(\frac{1}{\sqrt{2}} \right)^3 \\ &= \frac{1}{6\sqrt{2}} \\ &= \frac{\sqrt{2}}{12} \end{aligned}$$

16. Calculus, EXT1 C2 2009 HSC 1f

$$u = x^3 + 1$$

$$\frac{du}{dx} = 3x^2$$

$$du = 3x^2 dx$$

If $x = 2$, $u = 9$

$x = 0$, $u = 1$

$$\begin{aligned} \therefore \int_0^2 x^2 e^{x^3+1} dx &= \frac{1}{3} \int_0^2 e^{x^3+1} \cdot 3x^2 dx \\ &= \frac{1}{3} \int_1^9 e^u du \\ &= \frac{1}{3} [e^u]_1^9 \\ &= \frac{1}{3} (e^9 - e) \end{aligned}$$

17. Calculus, EXT1 C2 2005 HSC 1d

$$u = 2x^2 + 1$$

$$\frac{du}{dx} = 4x$$

$$du = 4x \, dx$$

$$dx = \frac{du}{4x}$$

$$\begin{aligned} \therefore \int x(2x^2 + 1)^{\frac{5}{4}} \, dx \\ &= \int x \cdot u^{\frac{5}{4}} \cdot \frac{du}{4x} \\ &= \frac{1}{4} \int u^{\frac{5}{4}} \, du \\ &= \frac{1}{4} \cdot \frac{4}{9} u^{\frac{9}{4}} + c \\ &= \frac{1}{9} u^{\frac{9}{4}} + c \\ &= \frac{1}{9} (2x^2 + 1)^{\frac{9}{4}} + c \end{aligned}$$

18. Calculus, EXT1 C2 2007 HSC 1e

$$u = 25 - x^2$$

$$\frac{du}{dx} = -2x$$

$$du = -2x \, dx$$

$$\text{If } x = 4, \, u = 9$$

$$x = 3, \, u = 16$$

$$\begin{aligned} \therefore \int_3^4 \frac{2x}{\sqrt{25 - x^2}} \, dx \\ &= - \int_{16}^9 u^{-\frac{1}{2}} \, du \\ &= - \left[\frac{1}{\frac{1}{2}} u^{\frac{1}{2}} \right]_{16}^9 \\ &= - [2\sqrt{u}]_{16}^9 \\ &= - [2\sqrt{9} - 2\sqrt{16}] \\ &= - [6 - 8] \\ &= 2 \end{aligned}$$

MARKER'S COMMENT: A "significant number" of students put the integral limits in the wrong order.

19. Calculus, EXT1 C2 2008 HSC 2a

$$u = \log_e x$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$\therefore du = \frac{1}{x} dx$$

$$\text{When } x = e^2, u = \log_e e^2 = 2$$

$$x = e, u = 1$$

$$\therefore \int_e^{e^2} \frac{1}{x(\log_e x)^2} dx$$

$$= \int_1^2 \frac{1}{u^2} du$$

$$= \int_1^2 u^{-2} du$$

$$= \left[-\frac{1}{u} \right]_1^2$$

$$= \left[\left(-\frac{1}{2} \right) - (-1) \right]$$

$$= \frac{1}{2}$$

WARNING: Most errors were made in the last stage of substitution in this question. *Be careful!*

20. Calculus, EXT1 C2 2013 HSC 11f

$$\text{Let } u = e^{3x}$$

$$\frac{du}{dx} = 3e^{3x}$$

$$\therefore dx = \frac{du}{3e^{3x}}$$

$$\text{When } x = \frac{1}{3}, u = e^{3 \times \frac{1}{3}} = e$$

$$x = 0, u = e^0 = 1$$

$$\therefore \int_0^{\frac{1}{3}} \frac{e^{3x}}{e^{6x} + 1} dx$$

$$= \int_1^e \frac{e^{3x}}{u^2 + 1} \times \frac{du}{3e^{3x}}$$

$$= \frac{1}{3} \int_1^e \frac{1}{u^2 + 1} du$$

$$= \frac{1}{3} [\tan^{-1} u]_1^e$$

$$= \frac{1}{3} [\tan^{-1} e - \tan^{-1} 1]$$

$$= \frac{1}{3} \left(\tan^{-1} e - \frac{\pi}{4} \right)$$

MARKER'S COMMENT: Many students did not calculate in radians and incorrectly got an answer of 8.2. *BE CAREFUL!* Note that converting your answer to 0.14 is also correct but not required.

$$x = u^2 + 1$$

$$u^2 = x - 1$$

$$u = \sqrt{x - 1}$$

$$\frac{du}{dx} = \frac{1}{2}(x - 1)^{-\frac{1}{2}}$$

$$= \frac{1}{2\sqrt{x - 1}}$$

$$\Rightarrow 2du = \frac{dx}{\sqrt{x - 1}}$$

When $x = 5$, $u = 2$

$x = 2$, $u = 1$

$$\therefore \int_2^5 \frac{x}{\sqrt{x - 1}} dx$$

$$= 2 \int_1^2 u^2 + 1 du$$

$$= 2 \left[\frac{u^3}{3} + u \right]_1^2$$

$$= 2 \left[\left(\frac{8}{3} + 2 \right) - \left(\frac{1}{3} + 1 \right) \right]$$

$$= \frac{20}{3}$$